

On speed estimation of permanent magnet synchronous motor using adaptive robust position observer and differentiator

Alaa HIJAZI*, Lilia SIDHOM*, Alois ZGORSKI**, Xuefang LIN SHI*

* *Laboratoire Ampère, UMR CNRS 5005, Université de Lyon, INSA-Lyon, bat Leonard de Vinci, F-69626 Villeurbanne, Cedex, France (Tel: +33 4 72 43 87 39; e-mail: alaa.hijazi@insa-lyon.fr).*

** *SAGEM DEFENSE SECURITE, 95, Route de Montélier, 26002 Valence sz France (Tel: +33 / (4) – 475867067; e-mail: alois.zgorski@sagem.com).*

Abstract: This article concerns a novel method for sensorless speed control of Permanent Magnet Synchronous Motor based on adaptive supertwisting sliding mode algorithm. This algorithm is used for motor position observation and for the derivation of the position to deduce the speed. The stability is deduced based on the Lyapunov theory. The Experimental tests, carried out on a test bench, show the pertinence of the proposed scheme.

1. INTRODUCTION

PMSM is gaining a lot of interests in industrial applications due to their high efficiency and power density. During the last decade, an importance has been considered to the field of sensorless control of motors because of several disadvantages of those sensors like the additional cost, extra space, volume and weight on the system (Nabti, et al., 2005) in addition to the weak reliability of the those elements.

In (Tomonobu, et al., 1995), the speed of the motor is obtained by calculating the flux linkage using integrator process. However, the main disadvantage of this method is coming from the drift caused by integrators.

Classical Luenberger observer was used in many works (Ben Hamed & Sbita, 2006; Grouz & Sbita, 2009; Bhangu & Bingham, 2003; Hamda, et al., 1999) to estimate the speed and the position of PMSM. A linearized feedback was used in (Hamda, et al., 1999) to make the design of the observer easier and similar to linear system.

Through experimental results, the effectiveness of Extended Kalman filter was shown in many works (Jang, et al., 2008; Vyncke, et al., 2010; Carriere, et al., 2010; Dhaouadi, et al., 1999). The main drawback of this algorithm is its long computation time due to the requirements to calculate at each time step the Jacobians and the hard initialization process (Jang, et al., 2008). In addition, the model in the EKF algorithm greatly depends on the motor's parameters and by consequence the parametric robustness of this algorithm is affected.

To overcome these drawbacks, the theory of sliding mode is used in (E. Urlep & Jezemik, 2007; Paponpen & Konghirun, 2006; Peixoto, et al., 1995). These observers are suffering from the chattering problem caused by the switching signal. The effect of chattering could be reduced using a low pass

filter which add a delay time and caused an offset on the observed position. In (Paponpen & Konghirun, 2006), the discontinuous function is replaced by a sigmoid function in the sliding observer to reduce chattering. In (Zaltni, et al., 2010), a second order sliding mode observer is used in the same optic. The major problem associated with this algorithm is the tuning of its gains to match convergence and good performance. The accuracy of this differentiator depends on the choice of these parameters (Smaoui, et al., 2005). It has been shown that a simple modification of the spectral content of the input can greatly affects the performance (Sidhom, et al., 2011). In fact, these parameters depend on the Lipschitz constant of the signal. This problem was resolved by adding adaptive laws to the basic algorithm (Sidhom, et al., 2008).

The main goal of this paper is to apply an adaptive supertwisting algorithm for sensorless speed control of PMSM motor. This algorithm is used for the motor position observation and also for deducing the speed of the motor by the derivation of the position via an adaptive supertwisting differentiator.

This paper is organized as following:

In the second section, we will present the mathematical model of the motor. The third section is dedicated to the theoretical study of the adaptive supertwisting observer and differentiator. The stability will be proved referring to Lyapunov theory of stability. The position observer is presented in the fourth section. In the fifth section, we will present the experimental results

2. Mathematical model of the PMSM motor

The usual mathematical model of the non-salient motor in the fixed reference (α, β) is given by following system of equations:

$$\begin{cases} V_\alpha = R_s I_\alpha + L_s \frac{dI_\alpha}{dt} - \omega_e \phi_f \sin(\theta_e) \\ V_\beta = R_s I_\beta + L_s \frac{dI_\beta}{dt} + \omega_e \phi_f \cos(\theta_e) \\ J \frac{d\omega_r}{dt} = \frac{3}{2} P \phi_f (\cos(\theta_e) I_\beta - \sin(\theta_e) I_\alpha) - f \omega_r - T \end{cases} \quad (1)$$

where V_α , V_β , I_α and I_β are respectively the stator voltages and currents in the α - β reference frame, R_s and L_s are the stator resistance and inductance, ϕ_f is the flux of the permanent magnet, J is the inertia of the rotor, P is the number of pole pairs, f is the viscous friction coefficient, T is a torque applied on the motor, θ_e is the electrical position, ω_e is the electrical speed and ω_r is the mechanical speed of the rotor.

The emf e_α and e_β are equal to:

$$\begin{cases} e_\alpha = -\omega_e \phi_f \sin(\theta_e) \\ e_\beta = \omega_e \phi_f \cos(\theta_e) \end{cases} \quad (2)$$

3. Adaptive supertwisting differentiator

In this section, we will present the adaptive supertwisting algorithm used in signal differentiator. An extension to the observation problem will be provided at the end of this section.

Let the input signal of differentiator $f(t)$ be a function defined on $[0, \infty[$ measurable in Lebesgues sense. This signal is considered as the sum of two terms:

$$f(t) = f_0(t) + \varepsilon(t) \quad (3)$$

$f_0(t)$ is an unknown base signal with the derivative having a lipshictz constant $C > 0$. $\varepsilon(t)$ is a bounded Lebesgue-measurable noise with unknown features. $\varepsilon(t)$ is smaller than ε which is sufficiently small.

Let $u(t)$ be the derivative of x (which is an auxiliary variable) so that :

$$\dot{x} = u(t) \quad (4)$$

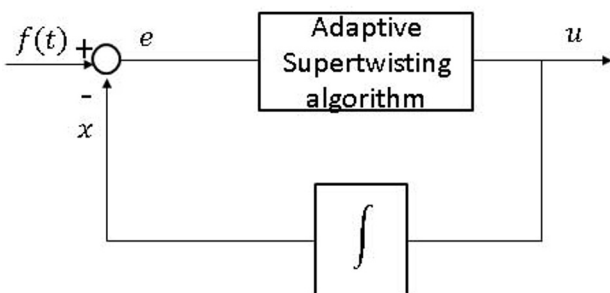


Fig. 1. Adaptive supertwisting algorithm structure

By applying a second order sliding algorithm (Levant, 1993) to keep $e = f(t) - x = 0$, (Levant, 1998) presents the following supertwisting differentiator:

$$u = \alpha \int \text{sgn}(e) dt + \gamma |e|^{\frac{1}{2}} \text{sgn}(e) \quad (5)$$

α and γ are constants.

$\text{Sgn}()$ is the sign function which is equal to :

$$\begin{cases} \text{sgn}(\epsilon) = 1 & \text{if } \epsilon > 0 \\ \text{sgn}(\epsilon) = -1 & \text{if } \epsilon < 0 \\ \text{sgn}(\epsilon) = [-1, 1] & \text{if } \epsilon = 0 \end{cases}$$

The sufficient conditions for the convergence of this algorithm are the following:

$$\alpha > C, \gamma^2 > 4C \frac{\alpha + C}{\alpha - C} \quad (6)$$

As we can see from (6), the choice of gains is closely depending on the Liptchiz constant C .

In the case of noised signal or when its spectral content changes, it is difficult to know a priori C . In addition, these gains shouldn't be chosen so high to not amplifying the noise at the output. All that makes difficult to have a good compromise between precision and robustness of this algorithm.

For easy adjustment of sliding mode differentiator's gains while providing a good compromise between precision and robustness, we propose to use an adaptive version of supertwisting algorithm which was proposed in (Sidhom, et al., 2011).

This equation presents the adaptive supertwisting differentiator:

$$u = \tilde{\alpha} \int \text{sgn}(e) dt + \tilde{\gamma} |e|^{\frac{1}{2}} \text{sgn}(e) + Ke \quad (7)$$

$\tilde{\alpha}$ and $\tilde{\gamma}$ are the adaptive gains, K is a constant.

Supposing that there exist two constants α^* and γ^* such that the following equation is verified:

$$\dot{f}(t) = \alpha^* \int \text{sgn}(e) dt + \gamma^* |e|^{\frac{1}{2}} \text{sgn}(e) \quad (8)$$

This expression is derived from the non-adaptive differentiator which was demonstrated to be convergent under certain conditions (6).

The derivative of the error e is equal to:

$$\dot{e} = \dot{f}(t) - u = -\hat{a} \int \text{sgn}(e) dt - \hat{\gamma} |e|^{\frac{1}{2}} \text{sgn}(e) - K_e e \quad (9)$$

Where $\hat{a}$ and $\hat{\gamma}$ are equal to:

$$\hat{a} = \tilde{a} - \alpha^* \quad (10)$$

$$\hat{\gamma} = \tilde{\gamma} - \gamma^* \quad (11)$$

Consider a Lyapunov candidate function V equal to :

$$V = \frac{1}{2} (e^2 + \hat{a}^2 + \hat{\gamma}^2) \quad (12)$$

The derivative of this Lyapunov function is equal to:

$$\dot{V} = e\dot{e} + \hat{a}\dot{\hat{a}} + \hat{\gamma}\dot{\hat{\gamma}} \quad (13)$$

$$= e \left[-\hat{a} \int \text{sgn}(e) dt - \hat{\gamma} |e|^{\frac{1}{2}} \text{sgn}(e) \right] - K_e e^2 + \hat{a}\dot{\hat{a}} + \hat{\gamma}\dot{\hat{\gamma}}$$

Given the following adaptive laws for $\hat{a}$ and $\hat{\gamma}$:

$$\dot{\hat{a}} = e |e|^{\frac{1}{2}} \text{sgn}(e) \quad (14)$$

$$\dot{\hat{\gamma}} = \tilde{\gamma} - \gamma^* \quad (15)$$

The Lyapunov function becomes:

$$\dot{V} = -K_e e^2 \leq 0 \quad (16)$$

So the differentiator is locally stable. However, we can only deduce that the differentiator error converges to zero ($e \rightarrow 0$). The convergence of $\hat{a}$ and $\hat{\gamma}$ toward zero couldn't be proved since $\dot{V}$ is equal to zero only when e tends to zero.

4. Adaptive supertwisting observer

Our goal is to observe the emf (e_α, e_β) of the motor and to deduce from it the position of the motor, given that:

$$\theta_e = \text{atan}\left(\frac{e_\alpha}{e_\beta}\right) \quad (17)$$

Considering the state vector composed of the stator current (I_α, I_β) and the EMF (e_α, e_β) in (α, β) reference frame. The state space representation of the motor is given by :

$$\dot{x} = \begin{bmatrix} \dot{I}_\alpha \\ \dot{I}_\beta \\ \dot{e}_\alpha \\ \dot{e}_\beta \end{bmatrix} = \begin{bmatrix} -\frac{R_s}{L_s} & 0 & -\frac{1}{L_s} & 0 \\ 0 & -\frac{R_s}{L_s} & 0 & -\frac{1}{L_s} \\ 0 & 0 & 0 & \omega_e \\ 0 & 0 & \omega_e & 0 \end{bmatrix} x + \begin{bmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix} \quad (18)$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} x$$

We can deduce from the calculation of observation matrix that it is full rank and by consequence the system is observable. Based on (2), we can deduce that the position of the motor is observable if the speed is non-zero.

The observer has the same structure as the differentiator; it is presented by the following equations:

$$\begin{cases} \frac{d\hat{I}_\alpha}{dt} = \frac{V_\alpha}{L_s} - \frac{R_s I_\alpha}{L_s} - \hat{e}_\alpha \\ \frac{d\hat{I}_\beta}{dt} = \frac{V_\beta}{L_s} - \frac{R_s I_\beta}{L_s} + \hat{e}_\beta \end{cases} \quad (19)$$

Where $\hat{e}_\alpha$ and $\hat{e}_\beta$ are equal to:

$$\hat{e}_i = \alpha_i \int \text{sgn}(e) + \gamma_i |e_i|^{\frac{1}{2}} \text{sgn}(e) + K_i e \quad (20)$$

The error term (e) is equal to the difference between the measured (I_i) and observed ($\hat{I}_i$) values. α_i , γ_i are the adaptive gains and K_i is a constant value.

The stability could be demonstrated by the same method as the differentiator.

5. Experimental results

To validate our approach, a test bench is developed. It consists of:

- A DC voltage source Xantrex (figure 3 (6))
- A commercial inverter 15 Kw (figure 3 (7)).
- A PMSM motor equipped with incremental coder of 4096 points by rotation (figure 2 (1)).
- A second motor (figure 3 (3)) to simulate the load piloted by a variable speed drive (figure 3 (4)).
- Three LEM current sensors (figure 3 (5)).
- DSPACE card (figure 3 (3)).

The characteristic of the motors are given in table I.

Table 1 . Motors characteristic.

Rated torque	5 Nm
Rated speed	3000 rpm
Rated voltage	540 V
R_s	2.06 Ω
L_s	9.15 mH
Φ_f	290 mWb
Number of pole pairs	3

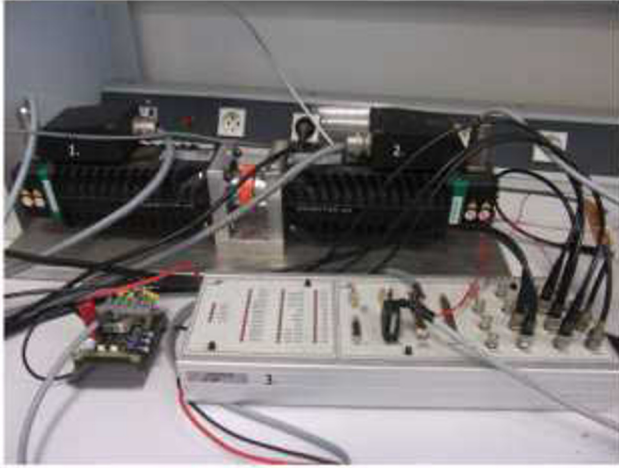


Fig. 2. Test bench : motor.



Fig. 3. Test bench power.

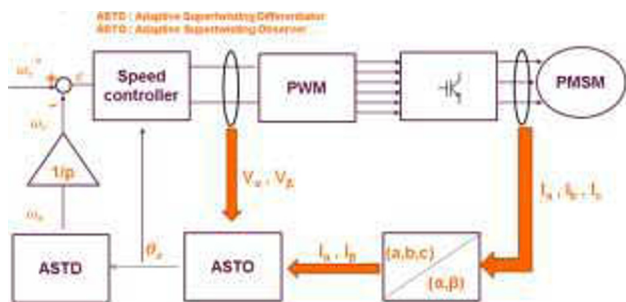


Fig. 4. Control structure.

The control algorithm (fig. 4) is performed by a dSpace ds1104 controller board, using Simulink and the Real-time workshop toolbox of MATLAB.

The parameters of supertwisting observer are initialized as following:

$$\begin{cases} \alpha_i=1000 \\ \gamma_i=1000 \\ K=400 \end{cases} \quad (21)$$

For the differentiator, the initial parameters are:

$$\begin{cases} \alpha=0 \\ \gamma=0 \\ K=20 \end{cases} \quad (22)$$

In order to examine the performance of the feedback loop (controller-observer), two tests was applied.

The first test consists of varying the speed reference of the control loop ($\omega_r^{\#}$). The speed of the motor was measured only for comparison with the observed one. In the figure 5, we present the measured and observed electrical speed of the motor as a function of time.

As we can see, the observed value converges toward the measured one very quickly and with excellent dynamic performance. We can also note that the speed of the motor converges toward the reference with no static error.

With the same test, the error estimation of electrical position is presented in figure 6. The error is less than 15 degree and it decreases below 5 degree when the speed is maximal.

In figure 7 and 8, we present the performance of the proposed method when a load variation is applied to test the robustness of the system. As we can see, within 0.5 seconds the regulation is insured. A constant electrical position estimation error of 10 degree is observed.

6. Conclusion

In this paper, we dealt with the problem of sensorless speed control of PMSM motor. The theory of adaptive supertwisting sliding mode was used in order to overcome the problem of parameters choices presented in classical supertwisting algorithm. Based on this algorithm, a position observer and differentiator was developed. Experimental results showed the pertinence of the proposed method. Further works could concentrate on the application of this method to zero speed application. It is also interesting to include observer and differentiator in the same algorithm by using higher order supertwisting algorithm

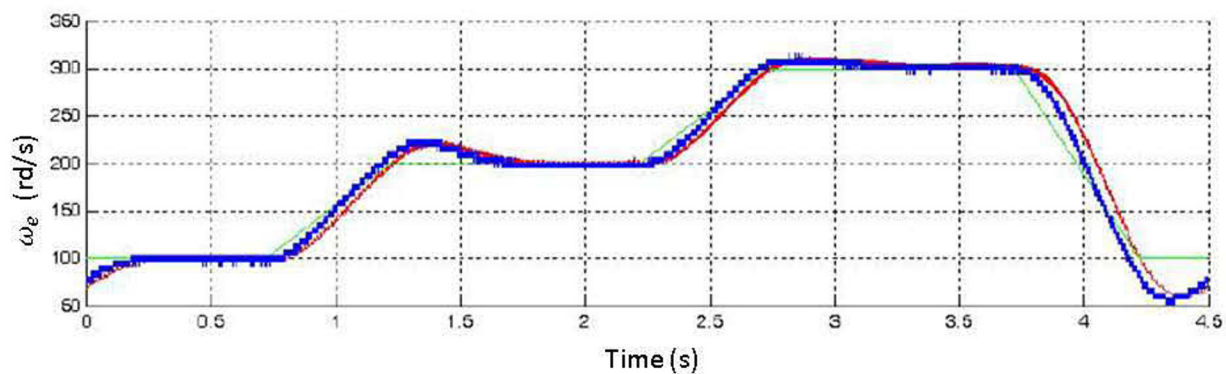


Fig. 5. Speed evolution: Green (reference), bleu (measured), red (observed).

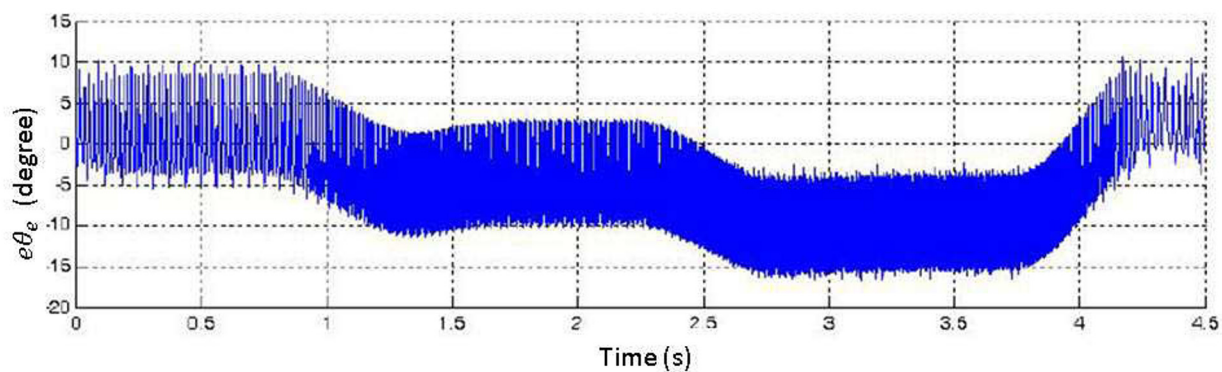


Fig. 6. Position error evolution

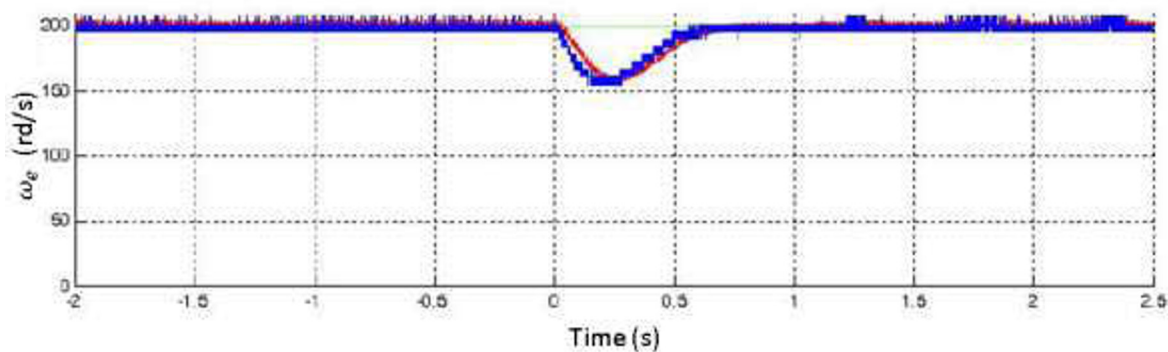


Fig. 7. Speed evolution toward load variation: Green (reference), bleu (measured), red (observed).

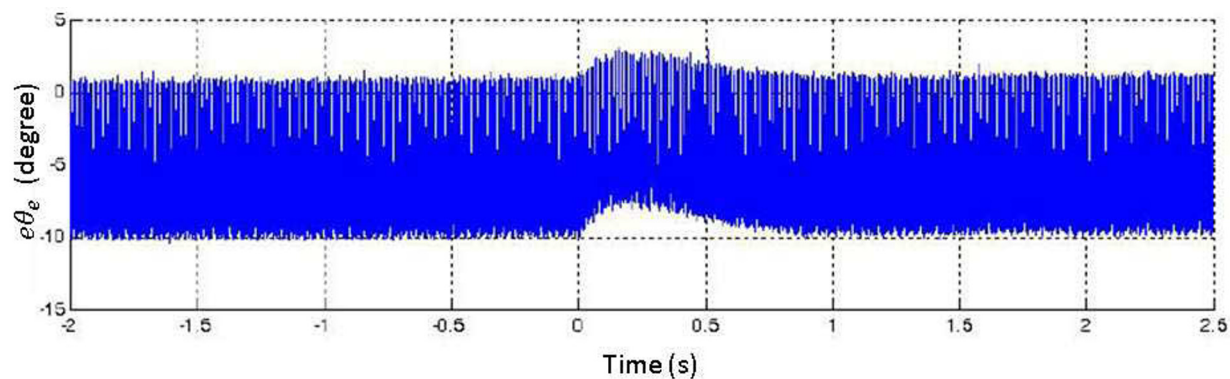


Fig. 8. Position error evolution.

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